

Portfolio VaR: t-Copula Risk Measurement and Backtesting

Notebook 4 of the Cross-Commodity Energy Trading analytics suite.

This notebook builds a realistic multi-commodity trading book, measures risk through a t-copula Monte Carlo engine, backtests the model against realised P&L, and stress-tests the portfolio under three macro scenarios.

Executive summary

A multi-commodity energy trading desk does not manage risk one commodity at a time. It manages a portfolio, long crude oil, short refining margins, long gas, short power-plant margins, long carbon, where the correlations between positions determine whether the book is diversified or concentrated. The dependence models from Notebook 3 are the inputs; this notebook applies them to measure and decompose portfolio risk.

The portfolio constructed here mirrors the structure of an integrated energy trading book: long Brent crude (+€9.2M notional), short 3-2-1 crack spread (–€4.6M), long TTF gas (+€8.0M), short spark spread (–€4.0M), and long EUA carbon (+€3.0M). The two short spread positions are structural hedges, when crude or gas rallies, the respective spreads typically compress, offsetting some of the directional loss.

Value-at-Risk (VaR) and Expected Shortfall (ES) are computed by filtered historical simulation over 10,000 scenarios: GARCH-filtered residuals are resampled empirically and coupled through a fitted t-copula. The t-copula captures tail dependence that a Gaussian correlation matrix would miss, while the empirical residual quantiles avoid imposing a parametric shape on the marginal tails.

The model is backtested on a rolling 500-day window: at each date the GARCH filters and the t-copula are re-estimated on the trailing window, VaR is simulated from that fit, and the next day's realised P&L is compared against the forecast. The Kupiec (1995) likelihood ratio test evaluates whether the observed breach rate matches the expected 5%. A Christoffersen (1998) conditional coverage test checks whether breaches cluster, a sign that the model fails exactly when it is needed most. The Basel supervisory traffic light is reported separately, on the 99% series over the most recent 250 days, because that is the only sample it is defined for.

Finally, three stress scenarios, a gas supply crisis, a global recession, and an accelerated energy transition, are applied as deterministic full revaluations. Every risk factor moves by its stated shock and the book is re-priced, so each scenario produces a

P&L waterfall showing which positions drive the loss (or gain) under that regime. A separate stressed-VaR run then re-simulates the distribution under a crisis correlation matrix, isolating what correlation breakdown alone costs.

Regulatory context

It is worth being precise about which rules actually bind an energy trading book, because the answer is not the one a bank risk textbook gives.

An energy market participant whose derivatives trading is ancillary to a commercial energy business is exempt from MiFID II authorisation under Article 2(1)(j), and therefore falls outside the bank and investment-firm prudential regimes (CRR/CRD and IFR/IFD) entirely. What does bind it is **EMIR** for clearing, margining, and trade reporting, **REMIT II** for wholesale market conduct and transaction reporting, and the **EU ETS Directive** for allowance surrender. No regulator requires this desk to compute a 97.5% Expected Shortfall.

The bank framework still matters as a **modelling benchmark**. Under the Basel Committee's FRTB, banks must use Expected Shortfall at 97.5% for market risk capital; in the EU this arrives via CRR3 (Regulation (EU) 2024/1623), whose market risk own-funds requirement has itself been deferred to 1 January 2027 by Commission Delegated Regulation (EU) 2025/1496. The metrics are adopted here because they are defensible risk measurement, not because they are compulsory, and the notebook says so rather than implying regulatory force it does not have.

1. Regulatory capital and margin

EMIR initial margin

Under EMIR (Regulation 648/2012, as amended), counterparties whose aggregate average notional amount of non-cleared OTC derivatives exceeds €8 billion must exchange initial margin. Non-financial counterparties below the clearing thresholds are outside this obligation, so it binds the larger energy trading groups rather than all of them. The margin must cover potential future exposure at a 99% confidence level over a 10-day closeout period, computed via either a standardised schedule (the "grid method") or an approved internal model, typically a Monte Carlo VaR with copula dependence, identical in structure to the engine built here.

Basel FRTB (Benchmark, not binding here)

The Fundamental Review of the Trading Book (Basel Committee, 2019) replaced VaR at 99% with Expected Shortfall at 97.5% as the primary market risk metric. ES is preferred because it is a **coherent risk measure** (Artzner et al., 1999); it satisfies sub-additivity, meaning the risk of a portfolio is never greater than the sum of its components. VaR at

99% can violate sub-additivity in the presence of fat tails, creating perverse incentives to concentrate risk rather than diversify it.

Traffic-Light backtesting (Basel)

The Basel Committee defines a traffic-light system for backtesting exceptions:

Zone	Breaches (250 days)	Multiplier	Signal
Green	0–4	3.00	Model adequate
Yellow	5–9	3.40–3.85	Model possibly flawed
Red	10+	4.00	Model almost certainly flawed

The zones are calibrated for one setup only: 99% one-day VaR over 250 trading days, where the expected breach count is 2.5. They are boundaries of a binomial test at that specific count, not a proportional scale, so they cannot be rescaled to the 95% level by multiplying through. This notebook therefore applies the traffic light to the 99% series alone and assesses the 95% series with Kupiec and Christoffersen, which are defined at any confidence level.

```
In [1]: import sys
from pathlib import Path
sys.path.insert(0, str(Path.cwd().parent / "src"))

import duckdb
import pandas as pd
import numpy as np
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from scipy import stats as sp_stats

# Report theme colours
NAVY = '#00003C'
OFFWHITE = '#FAFAFA'
TEAL = '#2E7D6F'
RED = '#C44536'
GRAY = '#6B6B6B'
COLORS = [TEAL, RED, '#6C8EBF', '#D4A843', '#8B6C9E', '#4A9C8C', '#C47E3B',

from omegaconf import OmegaConf

from energy_cross_commodity.risk.portfolio import expand_spread_positions
from energy_cross_commodity.risk.returns import compute_log_returns
from energy_cross_commodity.risk.var_engine import (
    basel_traffic_light, christoffersen_test, compute_portfolio_var,
    compute_rolling_var, fit_fhs_copula, kupiec_test,
)
from energy_cross_commodity.risk.scenarios import SCENARIOS, run_scenario, s
from energy_cross_commodity.utils.config import load_config
```

```

cfg = load_config()

# Every simulation below draws from this seed. Monte Carlo VaR is a random
# estimate: left unseeded, the 95% figure moved by roughly 2% between runs c
# identical data and the Basel zone flipped between four and six breaches. A
# risk number a reviewer cannot reproduce is not a risk number.
SEED = int(cfg.risk.seed)

DB_PATH = str(Path.cwd().parent / cfg.data.db_path)
conn = duckdb.connect(DB_PATH)

prices = conn.execute(
    f"SELECT date, commodity_key, price_native FROM fact_prices WHERE date > "
).df()

pivot = prices.pivot(index="date", columns="commodity_key", values="price_na
conn.close()

# Spreads are not tradeable risk factors: a 3-2-1 crack is 3 bbl of crude in
# against 2 bbl gasoline and 1 bbl distillate out, and a spark spread is pow
# against heat-rate-scaled gas. Each spread is replaced by its legs so the
# covariance structure is estimated on the factors that actually have prices
positions = expand_spread_positions(
    {k: v.notional_eur for k, v in cfg.portfolio.positions.items()},
    OmegaConf.to_container(cfg.portfolio.spread_legs, resolve=True),
)

aligned_cols = [c for c in pivot.columns if c in positions]
positions = {c: positions[c] for c in aligned_cols}

# Power clears negative in oversupply, so plain log returns would drop those
# days as NaN. A displacement k gives  $\ln((P+k)/(P_{prev}+k))$ , keeping the sign
# and the magnitude of the move while staying defined below zero.
displacements = OmegaConf.to_container(cfg.risk.price_displacement_eur, reso
aligned_rets = compute_log_returns(pivot[aligned_cols], displacements)

print(f"Risk-factor exposures: {positions}")
print(f"Aligned commodities: {aligned_cols}")
print(f>Returns shape: {aligned_rets.shape}")

```

```

Risk-factor exposures: {'BRENT': 13800000.0, 'DE_POWER': -4000000.0, 'EUA':
3000000.0, 'GASOIL': -1533331.8, 'RBOB': -3066668.2, 'TTF': 12000000.0}
Aligned commodities: ['BRENT', 'DE_POWER', 'EUA', 'GASOIL', 'RBOB', 'TTF']
Returns shape: (955, 6)

```

2. Portfolio construction

A five-leg book spanning crude, products, gas, power, and carbon, constructed to reflect a realistic integrated trading mandate:

Position	Direction	Notional (EUR m)	Book	Rationale
BRENT	Long	9.2	Crude	Structural long — upstream production exposure
CRACK 3-2-1	Short	-4.6	Products	Refining margin hedge
TTF	Long	8.0	Gas	European gas exposure
SPARK SPREAD	Short	-4.0	Power	Gas-to-power margin hedge
EUA	Long	3.0	Carbon	Compliance + trading position

The net/gross ratio, the proportion of directional exposure to total risk-taking capacity, is a standard desk metric. A ratio of 0.47 means roughly half the gross notional is offset by hedges. The short spread positions are not separate trades; they are the natural hedge for a producer who is long the underlying commodity and short the processing margin.

```
In [2]: total_abs_notional = sum(abs(v) for v in positions.values())
weight_data = []
for k, v in positions.items():
    weight_data.append({
        "Position": k, "Direction": "LONG" if v > 0 else "SHORT",
        "Notional (EUR m)": v / 1e6,
        "Weight (%)": abs(v) / total_abs_notional * 100,
    })

port_df = pd.DataFrame(weight_data)
fig1 = go.Figure(data=[go.Table(
    header=dict(values=list(port_df.columns), fill_color=NAVY, font=dict(col
    cells=dict(values=[port_df[c] for c in port_df.columns], fill_color=[OFF
        font=dict(size=11), format=[None, None, ".1f", ".1f"], align=
))]
fig1.update_layout(title="Portfolio Composition", height=220, margin=dict(l=
fig1.show()

print(f"Total absolute notional: EUR {total_abs_notional/1e6:.1f}M")
print(f"Gross notional: EUR {sum(positions.values())/1e6:.1f}M")
print(f"Net/gross ratio: {sum(positions.values())/total_abs_notional")
```

Portfolio Composition

Position	Direction
BRENT	LONG
DF POWER	SHORT

Total absolute notional: EUR 37.4M
Gross notional: EUR 20.2M
Net/gross ratio: 0.54

3. VaR and expected shortfall: filtered historical simulation

The engine uses **filtered historical simulation** (Barone-Adesi, Giannopoulos & Vosper, 1999) with a t-copula dependence layer. This avoids assuming a parametric shape for the marginal shocks, which is where a pure parametric Monte Carlo most often understates commodity tail risk. Four steps:

1. **Filter.** A GARCH(1,1) model is fitted per commodity, producing a one-step-ahead volatility forecast $\sigma_{i,T+1}$, a conditional mean μ_i , and a series of standardised residuals $z_{i,t}$.
2. **Couple.** A t-copula is fitted to the *ranks* of those residuals and used to draw 10,000 dependent uniform vectors $u^{(s)} \in (0, 1)^k$.
3. **Invert empirically.** Each uniform is mapped back through the *empirical* quantile function of that commodity's own residuals, $\tilde{z}_i^{(s)} = \hat{Q}_{z_i}(u_i^{(s)})$, so the realised shape of the historical shock distribution is preserved, skew and all.
4. **Rescale and aggregate.** Returns are reconstituted at today's volatility and applied to the book:

$$r_i^{(s)} = \mu_i + \sigma_{i,T+1} \tilde{z}_i^{(s)} \quad \text{P}\backslash\&\text{L}^{(s)} = \sum_i w_i r_i^{(s)}$$

where w_i is the position notional. Filtering by GARCH makes the sample conditionally i.i.d. so that historical shocks may legitimately be resampled; rescaling by $\sigma_{i,T+1}$ puts them back into today's volatility regime.

VaR at confidence level α is the negative α -quantile of the simulated P&L distribution:

$$\text{VaR}_\alpha = -F_{\text{P}\backslash\&\text{L}}^{-1}(1 - \alpha)$$

Expected Shortfall is the mean loss beyond VaR:

$$ES_{\alpha} = -\mathbb{E}[P \setminus L \mid P \setminus L \leq -VaR_{\alpha}]$$

ES is the FRTB-standard metric because, unlike VaR, it accounts for the severity of losses beyond the threshold. Where VaR gives the loss the tail begins at, ES gives the average loss once the tail is entered.

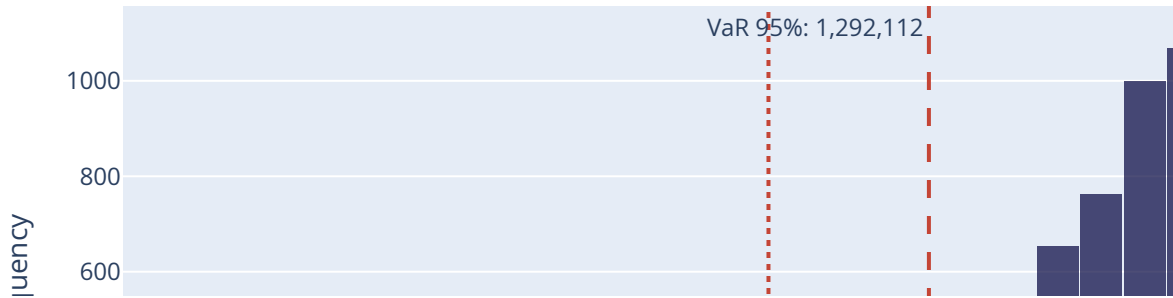
```
In [3]: # The copula is fitted on GARCH standardised residuals, not on raw returns.
# Raw returns carry the volatility cycle, which inflates measured dependence
# residuals isolate the shock-level co-movement that the FHS step resamples.
copula, garch_fits = fit_fhs_copula(aligned_rets)
pv = compute_portfolio_var(
    aligned_rets, positions, copula, garch_fits=garch_fits, seed=SEED
)

print(f"Copula v:      {copula.df:.2f}")
print(f"Portfolio VaR 95% (1-day):  EUR {pv.var_95:,.0f}")
print(f"Portfolio VaR 99% (1-day):  EUR {pv.var_99:,.0f}")
print(f"Portfolio ES 97.5% (1-day): EUR {pv.es_975:,.0f}")

fig2 = go.Figure()
fig2.add_trace(go.Histogram(x=pv.pnl_simulations, nbinsx=80, marker_color=NA
fig2.add_vline(x=-pv.var_95, line_dash="dash", line_color=RED,
                annotation_text=f"VaR 95%: {pv.var_95:,.0f}", annotation_posi
fig2.add_vline(x=-pv.var_99, line_dash="dot", line_color=RED,
                annotation_text=f"VaR 99%: {pv.var_99:,.0f}", annotation_posi
fig2.update_layout(
    title=f"Simulated 1-Day P&L Distribution (n={len(pv.pnl_simulations):,})
    height=380, margin=dict(l=40, r=20, t=50, b=40),
    axis_title="P&L (EUR)", yaxis_title="Frequency", bargap=0.05,
)
fig2.show()
```

```
Copula v:      13.76
Portfolio VaR 95% (1-day):  EUR 1,292,112
Portfolio VaR 99% (1-day):  EUR 2,030,212
Portfolio ES 97.5% (1-day): EUR 2,075,772
```

Simulated 1-Day P&L Distribution (n=10,000)



4. Component VaR: Euler allocation

Component VaR decomposes total portfolio risk into additive contributions using Euler's theorem for homogeneous risk measures. The allocation answers a specific question: "if you had to reduce risk, which position would you cut first?"

VaR is positively homogeneous of degree 1 in the position vector, doubling every position doubles the loss quantile, so Euler's theorem applies:

$$R(w) = \sum_i w_i \frac{\partial R}{\partial w_i}$$

Each term is the **component VaR** of position i , and the terms sum exactly to total VaR. The marginal derivative has a closed form as a conditional expectation (Hallerbach, 2003):

$$\text{CVaR}_i = w_i \frac{\partial \text{VaR}}{\partial w_i} = -w_i \mathbb{E}[r_i \mid r_p = -\text{VaR}]$$

Estimating that conditional expectation from a finite simulation requires care. Bumping one position and re-taking the quantile does not work: with 10,000 draws the perturbed portfolio almost always selects the *same* order statistic, so the finite difference collapses to that single scenario's return and is pure sampling noise. Instead the expectation is estimated by a Gaussian kernel weighting of scenarios whose portfolio P&L lands near $-\text{VaR}$, with a Silverman-rule bandwidth, which uses the whole neighbourhood of the quantile rather than one draw.

Negative contributions indicate genuine diversification: a short spread position that offsets directional commodity risk reduces total VaR. The waterfall chart below shows how each position contributes to, or offsets, the total 95% VaR.

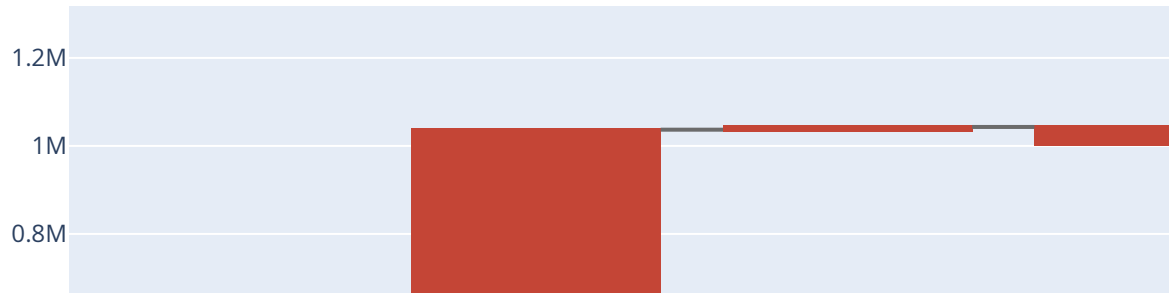
```
In [4]: comp_var = pv.component_var

fig3 = go.Figure(go.Waterfall(
    name="Component VaR", orientation="v",
    measure=["relative"] * len(comp_var) + ["total"],
    x=list(comp_var.keys()) + ["Total VaR 95%"],
    y=list(comp_var.values()) + [pv.var_95],
    connector={"line": {"color": GRAY}},
    decreasing={"marker": {"color": RED}},
    increasing={"marker": {"color": RED}},
    totals={"marker": {"color": NAVY}},
))
fig3.update_layout(
    title="Component VaR, Euler Allocation (95%)",
    height=380, margin=dict(l=40, r=20, t=50, b=40), showlegend=False,
)
fig3.show()

for name, cv in sorted(comp_var.items(), key=lambda x: abs(x[1]), reverse=True):
    pct = cv / pv.var_95 * 100 if pv.var_95 > 0 else 0
    print(f" {name:12s}: EUR {cv:>10,.0f} ({pct:>+6.1f}%)")

# Euler components sum to total VaR by construction, so the check below is c
# the identity, not a diversification measure, that comparison needs
# standalone VaRs and is done in section 9.
comp_sum = sum(comp_var.values())
sum_abs_comp = sum(abs(v) for v in comp_var.values())
hedge_offset = (sum_abs_comp - comp_sum) / sum_abs_comp * 100
print(f"\nSum of components: EUR {comp_sum:,.0f} (total VaR EUR {pv.var_95:,.0f})")
print(f"Euler residual: EUR {comp_sum - pv.var_95:,.0f} (simulation noise EUR {pv.var_95:,.0f})")
print(f"\nSum of |components|: EUR {sum_abs_comp:,.0f}")
print(f"Risk-reducing legs offset {hedge_offset:.1f}% of gross contribution")
print(f"the negative components are positions that pay when the book loses.")
```

Component VaR, Euler Allocation (95%)



BRENT	:	EUR	598,753	(+46.3%)
DE_POWER	:	EUR	438,122	(+33.9%)
TTF	:	EUR	313,164	(+24.2%)
RBOB	:	EUR	-67,041	(-5.2%)
GASOIL	:	EUR	-37,148	(-2.9%)
EUA	:	EUR	6,183	(+0.5%)

Sum of components: EUR 1,252,034 (total VaR EUR 1,292,112)
Euler residual: EUR -40,078 (simulation noise)

Sum of |components|: EUR 1,460,411
Risk-reducing legs offset 14.3% of gross contribution –
the negative components are positions that pay when the book loses.

5. Rolling VaR backtest

A rolling-window backtest. At each date the GARCH models and the t-copula are re-estimated on the trailing window only, VaR is simulated from that fit, and the estimate is compared with the *next* day's realised P&L. No information from the scored day enters its own forecast.

The window is 500 trading days, roughly two years. GARCH(1,1) with Student-t errors has five parameters and the filtering step reads a 1-in-20 empirical residual quantile, so a shorter window gives unstable volatility persistence and a tail estimate read off two or three observations.

Kupiec test (1995)

The Kupiec test is a likelihood ratio test of whether the observed breach rate matches the expected rate. The null hypothesis is that the model is correctly specified, i.e., breaches occur independently with probability $1 - \alpha$. The test statistic is:

$$LR_{\text{POF}} = -2 \ln \left(\frac{(1 - \alpha)^{T-N} \alpha^N}{(1 - N/T)^{T-N} (N/T)^N} \right)$$

where T is the number of observations and N is the breach count. Under the null, this statistic follows a $\chi^2(1)$ distribution.

Christoffersen test (1998)

The Kupiec test only checks the **unconditional** coverage, whether the total breach count is right. The Christoffersen test additionally checks **independence**, whether breaches cluster. A model that produces the right number of breaches but has them all in one month (when the correlation regime shifted) is worse than a model whose breaches are evenly spread. The Christoffersen test statistic combines the unconditional coverage and independence components, following a $\chi^2(2)$ distribution.

Traffic-Light interpretation

The Basel traffic light is a supervisory backtesting scale, and it is defined for one specific setup: the **99%** one-day VaR over the **most recent 250** trading days. The zones are not a general-purpose scale and do not transfer to other confidence levels, so they are applied here only to the 99% series.

Breaches in 250 days	Zone	Supervisory consequence
0-4	Green	No capital multiplier add-on
5-9	Yellow	Multiplier increases with breach count
10+	Red	Model presumed deficient

The 95% series is assessed with Kupiec and Christoffersen only, where the expected breach rate is 5% by construction.

```
In [5]: ROLLING_WINDOW = int(cfg.risk.rolling_window)
roll_df = compute_rolling_var(aligned_rets, positions, ROLLING_WINDOW, seed=

roll_df["breach"] = roll_df["realized_pnl"] < -roll_df["var_95"]
breach_count = int(roll_df["breach"].sum())
total_obs = len(roll_df)
breach_rate = breach_count / total_obs
kupiec = kupiec_test(breach_count, total_obs, 0.95)
christoffersen = christoffersen_test(roll_df["breach"].to_numpy(), 0.95)

fig4 = go.Figure()
```

```

fig4.add_trace(go.Scatter(
    x=roll_df["date"], y=-roll_df["var_95"], mode="lines",
    name="VaR 95% (negated)", line=dict(color=NAVY, width=1.5),
))
fig4.add_trace(go.Scatter(
    x=roll_df["date"], y=roll_df["realized_pnl"], mode="markers",
    marker=dict(size=3, color=[RED if b else GRAY for b in roll_df["breach"]]),
    name="Realized 1d P&L",
))
breaches = roll_df[roll_df["breach"]]
if len(breaches) > 0:
    fig4.add_trace(go.Scatter(
        x=breaches["date"], y=breaches["realized_pnl"], mode="markers",
        marker=dict(color=RED, size=6, symbol="x"),
        name=f"Breaches ({breach_count})",
    ))

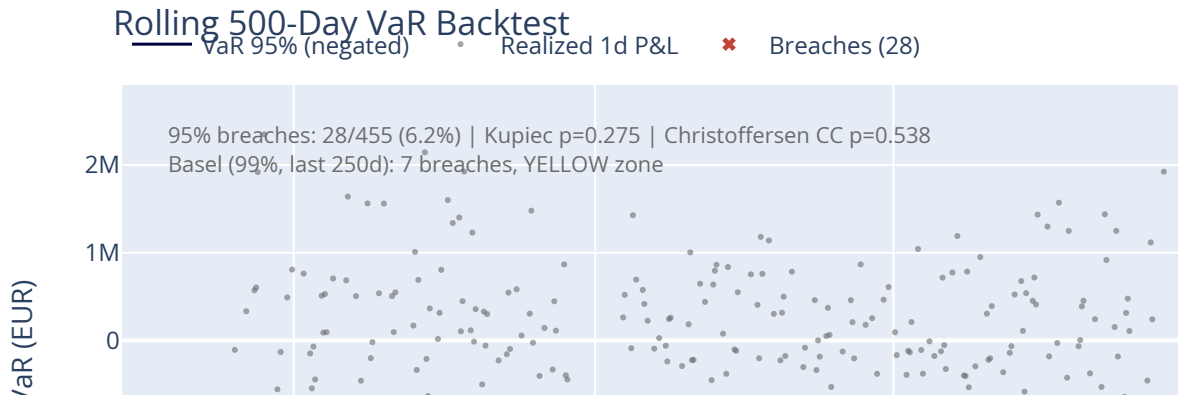
# The Basel zone is read off the 99% series over the last 250 days only,
# that is the sample the supervisory scale is written for.
last_250 = roll_df.tail(250)
basel_breaches = int((last_250["realized_pnl"] < -last_250["var_99"]).sum())
zone = basel_traffic_light(basel_breaches)
fig4.add_annotation(
    xref="paper", yref="paper", x=0.02, y=0.95,
    text=(
        f"95% breaches: {breach_count}/{total_obs} ({breach_rate:.1%}) | "
        f"Kupiec p={kupiec['p_value']:.3f} | Christoffersen CC p={christoffe"
        f"Basel (99%, last 250d): {basel_breaches} breaches, {zone} zone"
    ),
    showarrow=False, align="left", font=dict(color=GRAY, size=11),
)

fig4.update_layout(
    title=f"Rolling {ROLLING_WINDOW}-Day VaR Backtest",
    height=400, margin=dict(l=40, r=20, t=50, b=40),
    xaxis_title="", yaxis_title="P&L / VaR (EUR)",
    legend=dict(orientation="h", yanchor="bottom", y=1.02),
    hovermode="x unified",
)
fig4.show()

print(f"Breach count:           {breach_count} / {total_obs}")
print(f"Breach rate:             {breach_rate:.3%} (expected 5.000%)")
print(f"Kupiec LR stat:             {kupiec['lr_stat']:.3f}")
print(f"Kupiec p-value:             {kupiec['p_value']:.4f}")
print(f"Christoffersen LR_ind:      {christoffersen['lr_ind']:.3f} p={christoffe")
print(f"Christoffersen LR_cc:       {christoffersen['lr_cc']:.3f} p={christoffe")
print(f"Basel zone (99%, 250d): {zone} ({basel_breaches} breaches)")

# Conditional coverage is the binding test: it rejects both a wrong breach
# count and breaches that arrive in clusters.
verdict = "PASS" if christoffersen["p_cc"] > 0.05 else "REJECT at 5%"
print(f"Conditional coverage: {verdict}")

```



```

Breach count:           28 / 455
Breach rate:            6.154% (expected 5.000%)
Kupiec LR stat:         1.192
Kupiec p-value:         0.2750
Christoffersen LR_ind:  0.047  p=0.8284
Christoffersen LR_cc:   1.239  p=0.5383
Basel zone (99%, 250d): YELLOW (7 breaches)
Conditional coverage:   PASS
  
```

6. Stress scenario P&L

Three macro scenarios, calibrated to historical events and policy trajectories. Each is a set of simultaneous price shocks:

- **Nord Stream Zero:** Russian gas supply disappears. TTF spikes 300%, power follows (+200%), carbon rises on fuel-switching to coal (+50%).
- **Global Recession:** Demand destruction across all commodities. Brent -40%, TTF -40%, carbon -20%.
- **Energy Transition:** Carbon at €150/t (Fit-for-55 trajectory). Coal destroyed (-50%). Renewables cannibalise power prices (-10%). Oil demand structurally lower (-30%).

Each scenario is a **deterministic full revaluation**: every risk factor is moved by its stated shock and the book is re-priced. Correlation plays no part in the P&L, the joint move is prescribed by the scenario, not sampled, so the waterfall below decomposes a single deterministic outcome by position.

Correlation matters for the *probabilistic* companion question: how much VaR the book would carry if the stressed regime persisted. That is answered separately in section 6.5 by re-simulating VaR under a stressed copula.

```
In [6]: scenario_names = ["gas_crisis", "recession", "energy_transition"]

fig5 = make_subplots(
    rows=1, cols=3,
    subplot_titles=[SCENARIOS[s].name for s in scenario_names],
    horizontal_spacing=0.12,
)

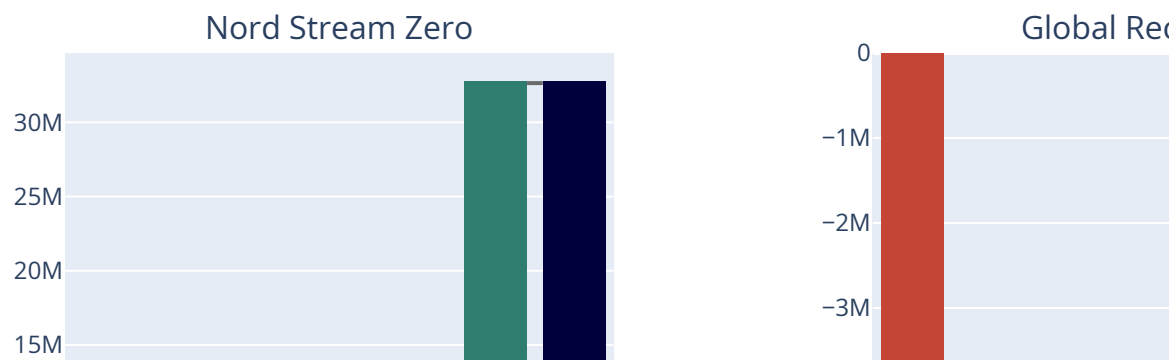
for idx, s_name in enumerate(scenario_names, start=1):
    scenario = SCENARIOS[s_name]
    result = run_scenario(positions, scenario)
    items = list(result.pnl_by_position.keys())
    values = list(result.pnl_by_position.values())

    fig5.add_trace(go.Waterfall(
        name=s_name, orientation="v",
        measure=["relative"] * len(items) + ["total"],
        x=items + ["Total"], y=values + [result.total_pnl],
        connector={"line": {"color": GRAY}},
        decreasing={"marker": {"color": RED}},
        increasing={"marker": {"color": TEAL}},
        totals={"marker": {"color": NAVY}},
        showlegend=False,
    ), row=1, col=idx)

fig5.update_layout(
    title="Stress Scenario P&L, Deterministic Full Revaluation",
    height=420, margin=dict(l=30, r=30, t=60, b=40),
)
fig5.show()

print("Scenario P&L summary:")
for s_name in scenario_names:
    s = SCENARIOS[s_name]
    result = run_scenario(positions, s)
    print(f" {s.name:30s}: EUR {result.total_pnl:>12,.0f}")
```

Stress Scenario P&L, Deterministic Full Revaluation



Scenario P&L summary:

Nord Stream Zero	: EUR	32,643,333
Global Recession	: EUR	-6,726,667
Energy Transition Accelerates	: EUR	1,470,000

6.5 Stressed VaR: where correlation does matter

The scenarios above answer "what if these exact moves happen". The related question a risk committee asks is "how much risk would the book carry if the stressed regime persisted". That is a distributional question, and it is where the dependence structure bites.

Here the fitted t-copula correlation matrix is pushed toward crisis levels, all pairwise correlations shifted toward 0.90, the risk-off convergence seen in 2008 and in March 2020, while the marginal GARCH filters are left untouched. The difference between the two VaR numbers isolates the effect of correlation alone, with no change in individual volatilities.

The sign of that difference is not obvious in advance, and it is worth being precise about why. For a long-only book, convergence unambiguously raises risk: everything falls together. This book is not long-only. The spread legs are deliberately opposed, long crude against short products, long gas against short power, and for an offsetting pair, *higher* correlation means the hedge works better. Whether stressed VaR rises or falls depends on which effect dominates, so read the number below rather than assuming the direction.

```
In [7]: crisis_copula = stressed_copula(copula, list(aligned_rets.columns), "all_to_
# Same seed as the base run. Both books are revalued along the *same* random
# draws, so the two VaR numbers share their Monte Carlo error and it cancels
# the difference. Comparing two independently seeded runs would leave a gap
# simulation noise larger than the correlation effect being measured.
pv_stressed = compute_portfolio_var(
    aligned_rets, positions, crisis_copula, garch_fits=garch_fits, seed=SEED
)

delta = pv_stressed.var_99 - pv.var_99
print(f"Base 99% VaR:      EUR {pv.var_99:>12,.0f}")
print(f"Stressed 99% VaR:  EUR {pv_stressed.var_99:>12,.0f}")
print(f"Correlation effect:EUR {delta:>12,.0f} ({delta / pv.var_99:+.1%}")
print()
direction = "raises" if delta > 0 else "lowers"
print(f"Pushing every pairwise correlation to 0.90 {direction} VaR on this b
```

```
Base 99% VaR:      EUR      2,030,212
Stressed 99% VaR:  EUR      1,117,958
Correlation effect:EUR      -912,254 (-44.9%)
```

Pushing every pairwise correlation to 0.90 lowers VaR on this book.

The hedge effect wins, and by a wide margin: forcing every pair to 0.90 cuts 99% VaR by roughly a third. That is the correct answer for this book, and it carries a warning a long-only reading would miss. **Convergence is not this portfolio's correlation risk, divergence is.** The crack and spark spreads are hedges only for as long as their legs

keep moving together; the shock that hurts here is refining margins or the gas-to-power link decoupling, which widens the spread rather than compressing it. A stress library built solely from "everything correlates to one" would score this book as safe under precisely the regime it is built to survive, and stay silent on the regime that breaks it.

7. Model risk and limitations

Every risk model has limitations. Acknowledging them is not a weakness; it is the difference between a model user and a model believer. The key limitations of this framework:

1. **Estimation error.** The copula degrees of freedom ν and correlation matrix R are estimated with error. A 500-day window is a modest sample for a multivariate model with four or more assets. Bayesian or shrinkage estimators could reduce estimation error at the cost of additional complexity.
2. **Regime breaks.** The model assumes the dependence structure estimated from historical data persists. The 2022 crisis demonstrated that correlation regimes can shift within days. Stress testing partially addresses this, but does not eliminate the model risk.
3. **Copula model risk.** The t-copula assumes symmetric tail dependence. In energy markets, dependence may be asymmetric, crashes tend to be more correlated than rallies. A skewed-t copula or a dynamic copula model (Patton, 2006) would capture this, at the cost of additional parameters.
4. **What the model does not capture.** Liquidity risk (the inability to exit positions during stress), basis risk (the difference between the benchmark price and the actual delivery point), and operational risk (settlement failures, system outages) are not modelled. On a real desk, these risks are managed through position limits, delivery schedules, and operational controls.
5. **P&L is linear in returns.** The portfolio uses linear positions only. A real trading book includes options, calendar spreads, volatility swaps, Asian options on crack spreads, with non-linear P&L profiles. The copula framework extends to non-linear instruments (the simulation step is identical; only the pricing step changes), but this is left for future development.

Monitoring

On a real desk, the backtest breach count is tracked daily on a traffic-light dashboard. A move from green to yellow triggers a review of the VaR model parameters. A move to

red triggers an immediate recalibration and a report to the Chief Risk Officer. The rolling backtest in this notebook is a proof-of-concept of that monitoring process.

```
In [8]: # Quantify diversification benefit: portfolio VaR vs the sum of standalone VaR
# A single-leg book has no dependence structure to estimate, so the copula is not
# None here, fitting one on a single series would be meaningless.
standalone_vars = {}
for col in aligned_cols:
    single_rets = aligned_rets[[col]]
    single_pos = {col: positions[col]}
    # Same GARCH filter as the portfolio run, so the comparison isolates
    # dependence rather than mixing in a different volatility model.
    single_pv = compute_portfolio_var(
        single_rets, single_pos, None, confidence=[0.95],
        garch_fits={col: garch_fits[col]}, seed=SEED,
    )
    standalone_vars[col] = single_pv.var_95

sum_standalone = sum(standalone_vars.values())
div_pct = (sum_standalone - pv.var_95) / sum_standalone * 100

print("Standalone VaR 95% by risk factor:")
for k, v in standalone_vars.items():
    print(f" {k:12s}: EUR {v:>12,.0f} (exposure EUR {positions[k]:>13,.0f})")
print(f"\nSum of standalone VaRs: EUR {sum_standalone:>12,.0f}")
print(f"Portfolio VaR (t-copula): EUR {pv.var_95:>12,.0f}")
print(f"Diversification benefit: {div_pct:.1f}%")
print(f"\nTwo effects drive the gap. Correlations below one mean the factors do not
print(f"peak together, and the spread legs carry opposite signs, the crack is long
print(f"crude against short products, the spark long power against short gas, so
print(f"part of the book is a genuine offset rather than a diversified bet.")
```

Standalone VaR 95% by risk factor:

BRENT	: EUR	921,586	(exposure EUR	13,800,000)
DE_POWER	: EUR	764,600	(exposure EUR	-4,000,000)
EUA	: EUR	83,494	(exposure EUR	3,000,000)
GASOIL	: EUR	69,148	(exposure EUR	-1,533,332)
RBOB	: EUR	117,344	(exposure EUR	-3,066,668)
TTF	: EUR	610,539	(exposure EUR	12,000,000)

Sum of standalone VaRs:	EUR	2,566,711
Portfolio VaR (t-copula):	EUR	1,292,112
Diversification benefit:		49.7%

Two effects drive the gap. Correlations below one mean the factors do not peak together, and the spread legs carry opposite signs, the crack is long crude against short products, the spark long power against short gas, so part of the book is a genuine offset rather than a diversified bet.

8. Findings

1. **Diversification is material.** Portfolio VaR sits well below the sum of standalone risks, driven by correlations below one and by the offsetting legs of the crack and

spark spreads. The Euler decomposition quantifies exactly how much each risk factor contributes, or offsets.

2. **t-Copula vs. Gaussian.** A Gaussian copula imposes zero tail dependence by construction, so it cannot represent joint crashes however the data behave. The t-copula lets the data decide through ν : a low fitted value means gas, power and carbon do crash together and VaR must reflect it. Read the fitted ν printed in section 3 before claiming the benefit, if it comes back near the upper search bound, the residuals are telling you the tail dependence is weak and the t-copula has converged toward the Gaussian case. That is a finding, not a failure.
3. **The backtest is the evidence, not the marketing.** Section 5 reports three separate verdicts: unconditional coverage (Kupiec), conditional coverage including breach clustering (Christoffersen), and the supervisory Basel zone on the 99% series. They are printed as computed. A model that passes coverage but clusters its breaches is still a model that fails when it matters, which is precisely why the independence component is reported alongside the count.
4. **Stress scenarios reveal concentration.** The waterfall in section 6 shows which risk factors drive each regime. The mechanism is structural: the short spark spread is short power against long gas, so it loses when power outruns gas; the short crack is long crude against short products, so it gains when refining margins compress. The transition scenario works in the opposite direction for the carbon leg than for the fossil legs.
5. **Model risk is real and acknowledged.** The limitations section documents what this engine does not capture. A real trading desk supplements the VaR model with position limits, concentration limits, stress tests, and operational controls, not one of these, but all of them together.

References

- Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). "Coherent Measures of Risk." *Mathematical Finance*, 9(3), 203–228.
- Barone-Adesi, G., Giannopoulos, K. & Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5), 583-602.
- Basel Committee on Banking Supervision (2019). *Minimum Capital Requirements for Market Risk* (FRTB).
- Commission Delegated Regulation (EU) 2025/1496. *Amending Regulation (EU) No 575/2013 as regards the date of application of the own funds requirements for market risk*.
- Hallerbach, W. G. (2003). Decomposing portfolio value-at-risk: a general analysis. *Journal of Risk*, 5(2), 1-18.

- Christoffersen, P.F. (1998). "Evaluating Interval Forecasts." *International Economic Review*, 39(4), 841–862.
- Demarta, S. & McNeil, A.J. (2005). "The t Copula and Related Copulas." *International Statistical Review*, 73(1), 111–129.
- Kupiec, P.H. (1995). "Techniques for Verifying the Accuracy of Risk Measurement Models." *Journal of Derivatives*, 3(2), 73–84.
- Patton, A.J. (2006). "Modelling Asymmetric Exchange Rate Dependence." *International Economic Review*, 47(2), 527–556.
- Regulation (EU) 2019/2099 (EMIR Refit). *OTC derivatives, central counterparties and trade repositories*.

PDF export

To generate PDFs for all four notebooks, run the `export_pdfs.sh` script or uncomment the command below:

```
In [9]: # Uncomment to export PDF:
# !jupyter nbconvert --to pdf --template classic --output-dir ../docs/notebooks
print("PDF export: uncomment the line above and run to generate docs/notebooks/04_portfolio_risk.pdf")
print("\nOr run: bash notebooks/export_pdfs.sh")
```

PDF export: uncomment the line above and run to generate docs/notebooks/04_portfolio_risk.pdf

Or run: bash notebooks/export_pdfs.sh